Matching for parameterised symmetric golden maps

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Outline

(i) Introduction

(ii) Matching

(iii) Invariant measures / frequencies of digits

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(i) Introduction

Symmetric golden maps

- Set $\beta := \frac{\sqrt{5}+1}{2}$ $(\beta^2 = \beta + 1)$, $J_0 := [-1/\beta, 1/\beta]$, $J_{\pm 1} := \pm(1/\beta, 1]$
- Let $s: [-1,1] \to \{0, \pm 1\}$, where $s(x) = i$ iff $x \in J_i$
- For $\alpha \in [1, \beta]$, define $S_\alpha : [-1,1] \to [-1,1]$ by

$$S_\alpha(x) := \beta x - s(x)\alpha$$

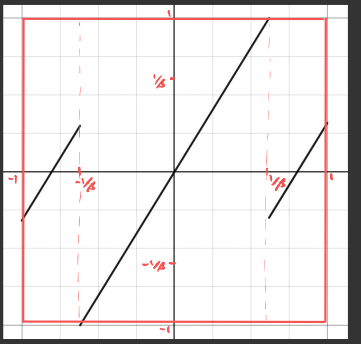
$$= \begin{cases} \beta x + \alpha, & x \in J_{-1} \\ \beta x, & x \in J_0 \\ \beta x - \alpha, & x \in J_1 \end{cases}$$

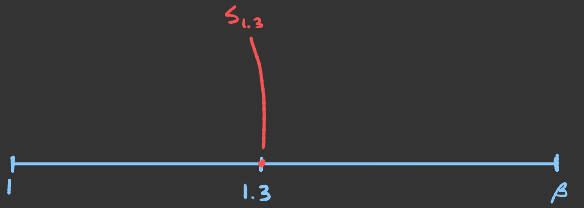
- Note $S_\alpha(J_1) = S_\alpha((1/\beta, 1])$

$$= (1-\alpha, \beta - \alpha]$$

$$\subset (-1/\beta, 1/\beta]$$

$$\subset J_0$$





S_α - expansions

- For $j \geq 1$, set $s_j = s_{\alpha,j}(x) := s\left(S_\alpha^{j-1}(x)\right)$
- Let $v : \{0, \pm 1\}^* \longrightarrow \mathbb{R}$
  $u = u_1 \cdots u_n \longmapsto v(u) = \sum_{j=1}^{n} u_j / \beta^j$
- Via induction, $S_\alpha^k(x) = \beta^k (x - \alpha v(s_1 \cdots s_k)) \quad \forall k \geq 0$. Hence

$$x = \alpha v(s_1 \cdots s_k) + S_\alpha^k(x)/\beta^k,$$

and letting $k \to \infty$,

$$x = \alpha \sum_{j \geq 1} s_j / \beta^j.$$

- Call $(s_j)_{j \geq 1} = (s_{\alpha,j}(x))_{j \geq 1}$ the S_α-expansion of x
- Since $S_\alpha(J_{\pm 1}) \subset J_0$, ± 1 is followed by 0

Main results

- **Thm. 1:** For each $\alpha \in (1, \beta]$, $\exists!$ abs. cont. S_α-invariant measure μ_α. In fact, $\mu_\alpha \sim \lambda$, and $\exists$ a countable collection $\{I_d\}_{d \in \mathcal{M}}$ of disjoint, open subintervals of $[1, \beta]$ with full λ-measure s.t. for fixed $d \in \mathcal{M}$, the density of each μ_α, $\alpha \in I_d$, is a step function with (at most) the same number of jumps

$$\left(\lim_{n \to \infty} \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{1}_{J_0}\left(S_\alpha^j(x)\right) \right)$$

- **Thm. 2:** The asymptotic relative frequency of $0^\vee$ in λ-a.e. S_α-expansion depends continuously on α and attains its max. $4/5$ on the maximal interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$

Prior work

- Dajani–Kalle (2020) obtain similar results for a family of maps defined analogously with constant slope 2

- Thm. 1: For each $\alpha \in (1, \cancel{\beta} 2]$, $\exists!$ abs. cont. S_α-invariant measure μ_α. In fact, $\mu_\alpha \sim \lambda$, and $\exists$ a countable collection $\{I_d\}_{d \in \mathcal{U}}$ of disjoint, open subintervals of $[1, \cancel{\beta} 2]$ with full λ-measure s.t. for fixed $d \in \mathcal{U}$, the density of each μ_α, $\alpha \in I_d$, is a step function with (at most) the same number of jumps

- Thm. 2: The asymptotic relative frequency of 0 in λ-a.e. S_α-expansion depends continuously on α and attains its max. ~~4/5~~ 2/3 on the maximal interval ~~$[1/2 + 1/\beta, 1 + 1/\beta^2]$~~ $[6/5, 3/2]$

Similarities	Differences
· Kopf (1990) · Matching · Birkhoff's Ergodic Thm.	· Dynamics of golden maps more subtle · DK20 relate binary maps to α-continued fractions · Our analysis more intrinsic to golden maps

(ii) Matching

Motivation and definitions

- Fix $\alpha \in (1, \beta]$. Results of K90 imply the existence of a unique — hence ergodic — abs. cont. S_α-invariant probability measure with density

$$f_\alpha(x) = \frac{1}{C} \sum_{j \geq 0} \sum_{a \in \{\pm 1/\beta\}} \frac{1}{\beta^{j+1}} \left(\mathbb{1}_{[-1,\, S_\alpha^j(\lim_{x \to a^-} S_\alpha(x))]}(x) - \mathbb{1}_{[-1,\, S_\alpha^j(\lim_{x \to a^+} S_\alpha(x))]}(x) \right)$$

with $C \in \mathbb{R}$ a normalising constant

- Note $\lim_{x \to 1/\beta^-} S_\alpha(x) = 1$ and $\lim_{x \to 1/\beta^+} S_\alpha(x) = 1 - \alpha$

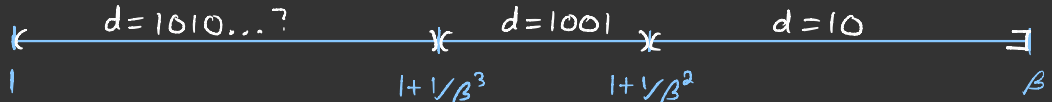
- Defn.: The matching index of S_α is $m = m(\alpha) := \inf_j \{ S_\alpha^j(1) = S_\alpha^j(1-\alpha) \}$; S_α has matching if $m < \infty$.

Let (d_j), (e_j) denote the S_α-expansions of 1, $1-\alpha$, resp., and set $d = d_1 \cdots d_m$, $e = e_1 \cdots e_m$. When $m < \infty$, call d the matching word corresponding to α. A maximal subinterval of $[1, \beta]$ on which matching words equal d is a matching interval

First examples

- Note $1 \in J_1$, $1-\alpha \in J_0$ $\forall \alpha \in [1,\beta]$, so $\begin{pmatrix} d_1 \\ e_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (throughout, $\overline{u} := -u$)

$\alpha \in (1+1/\beta^2, \beta]$	$S_\alpha(1) = \beta - \alpha \in [0, 1/\beta^3) \subset J_0$, $S_\alpha(1-\alpha) = \beta - \beta\alpha \in [-1,-1/\beta) \subset J_{-1}$, $S_\alpha^2(1) = \beta^2 - \beta\alpha \in J_0$ and $S_\alpha^2(1-\alpha) = \beta^2 - \beta^2\alpha + \alpha = \beta^2 - \beta\alpha \in J_0$	$\begin{pmatrix} d \\ e \end{pmatrix} = \begin{pmatrix} 10 \\ 01 \end{pmatrix}$
$\alpha = 1+1/\beta^2$	$S_\alpha(1) = \beta - \alpha = 1/\beta^3 \in J_0$, $S_\alpha(1-\alpha) = \beta - \beta\alpha = -1/\beta \in J_0$, $S_\alpha^2(1) = 1/\beta^2 \in J_0$, $S_\alpha^2(1-\alpha) = -1 \in J_{-1}$, $S_\alpha^3(1) = 1/\beta \in J_0$, $S_\alpha^3(1-\alpha) = -1/\beta^3 \in J_0$, $S_\alpha^4(1) = 1 \in J_1$ and $S_\alpha^4(1-\alpha) = -1/\beta^2 = 1-\alpha \in J_0$,	No matching (but a Markov partition)
$\alpha \in (1+1/\beta^3, 1+1/\beta^2)$	$S_\alpha(1) = \beta - \alpha \in (1/\beta^3, 1/\beta^2) \subset J_0$, $S_\alpha(1-\alpha) = \beta - \beta\alpha \in (-1/\beta, -1/\beta^2) \subset J_0$, $S_\alpha^2(1) = \beta^2 - \beta\alpha \in (1/\beta^2, 1/\beta) \subset J_0$, $S_\alpha^2(1-\alpha) = \beta^2 - \beta^2\alpha \in (-1,-1/\beta) \subset J_{-1}$, $S_\alpha^3(1) = \beta^3 - \beta^2\alpha \in (1/\beta, 1) \subset J_1$, $S_\alpha^3(1-\alpha) = \beta^3 - (\beta^3-1)\alpha \in (-1/\beta^3, 1/\beta^3) \subset J_0$, $S_\alpha^4(1) = \beta^4 - (\beta^3+1)\alpha \in J_0$ and $S_\alpha^4(1-\alpha) = \beta^4 - (\beta^4-\beta)\alpha \in J_0$.	$\begin{pmatrix} d \\ e \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$
$\alpha = 1+1/\beta^3$	$S_\alpha(1) = \beta - \alpha = 1/\beta^2 \in J_0$, $S_\alpha(1-\alpha) = \beta - \beta\alpha = -1/\beta^2 \in J_0$, $S_\alpha^2(1) = 1/\beta \in J_0$, $S_\alpha^2(1-\alpha) = -1/\beta \in J_0$, $S_\alpha^3(1) = 1 \in J_1$, $S_\alpha^3(1-\alpha) = -1 \in J_{-1}$ and $S_\alpha^4(1-\alpha) = -1/\beta^2 \in J_0$,	No matching (MP)
$\alpha = 1$	$S_\alpha(1) = 1/\beta \in J_0$, $S_\alpha^2(1) = 1 \in J_1$ and $S_\alpha(1-\alpha) = 0 = 1-\alpha \in J_0$.	No matching (MP)



Orbit differences

- Prop. 1: For every $\alpha \in [1, \beta]$ and $j \geq 0$,

$$\gamma_j := S_\alpha^j(1) - S_\alpha^j(1-\alpha) \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$$

- Sketch: Suppose $\alpha \in (1, 1+1/\beta^3)$. Note $\gamma_0 = \alpha$. Suppose $\gamma_{j-1} \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$.

If $\gamma_{j-1} = 0$, done, so suppose $\gamma_{j-1} \neq 0$. Note

$$\begin{aligned}\gamma_j &= \beta S_\alpha^{j-1}(1) - d_j\alpha - (\beta S_\alpha^{j-1}(1-\alpha) - e_j\alpha) \\ &= \beta\gamma_{j-1} - \alpha(d_j - e_j).\end{aligned}$$

Consider cases:

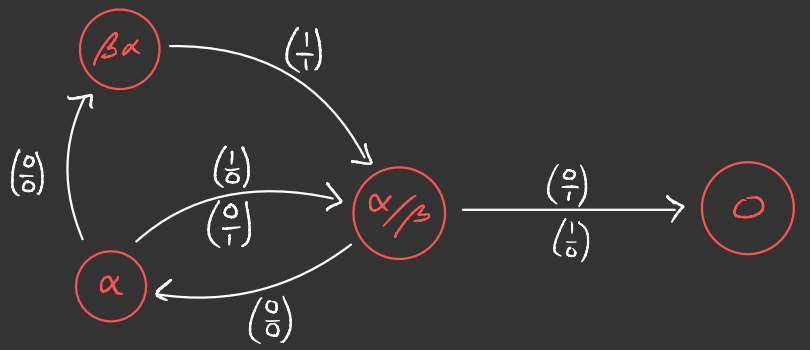
(i) $\gamma_{j-1} = \alpha/\beta$: Since $1/\beta < \alpha/\beta < 2/\beta$, $\begin{pmatrix} d_j \\ e_j \end{pmatrix} \in \left\{\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$.



Hence $\gamma_j \in \{0, \alpha\}$.

⋮

Orbit differences



Vertices: $y_{j-1} = S_\alpha^{j-1}(1) - S_\alpha^{j-1}(1-\alpha)$

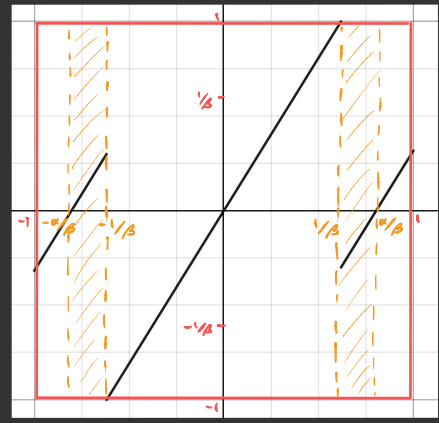
Edges: $\xrightarrow{\binom{d_j}{e_j}}$

□

Note $\begin{pmatrix} d_{m-2} & d_{m-1} & d_m \\ e_{m-2} & e_{m-1} & e_m \end{pmatrix} \in \left\{ \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right) \right\}$.

Conversely, if $\begin{pmatrix} d_{j-2} & d_{j-1} & d_j \\ e_{j-2} & e_{j-1} & e_j \end{pmatrix} \in \left\{ \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right) \right\}$, then $j = m$.

Matching almost everywhere



- Note that $\operatorname{diam}(J_{\pm 1}) = 1 - 1/\beta = 1/\beta^2 < \alpha/\beta$. By Prop. 1, $S_\alpha^j(1-\alpha) \in J_1$ or $S_\alpha^j(1) \in J_{-1}$ implies matching. But $S_\alpha(x)\big|_{J_0} = \beta x$, so $S_\alpha^j(1-\alpha) > 0$ or $S_\alpha^j(1) < 0$ implies matching. Hence

(*)
$$S_\alpha^j(1-\alpha) \in S_\alpha^{-1}((0,1]) \cap [-1,0) = (-\alpha/\beta, -1/\beta) \quad \text{or}$$
$$S_\alpha^j(1) \in S_\alpha^{-1}([-1,0)) \cap (0,1] = (1/\beta, \alpha/\beta)$$

implies matching.

- Prop.: S_α has matching for λ-a.e. $\alpha \in [1,\beta]$.

- Idea: Let $\ell_\alpha = \inf\{j \mid (*) \text{ holds}\}$, $\alpha > 1$. Show for $x \in [1, \alpha-1]$ and $0 \le j \le \ell_\alpha$, $S_\alpha^j(x) = \alpha B^j(x/\alpha)$, $B(x) = \beta x \pmod 1$.

Use ergodicity of B. □

Matching words/intervals

- Set $w_0 := 00$, $w_1 := 001$, $w_2 := 01$

- Defn.: Say $d \in \{0,1\}^*$ is in admissible block form if $d = 10$, 1001, or
$$d = 1 w_{i_1} \cdots w_{i_n} (1 - i_n/2), \quad i_1, \ldots, i_n \in \{0,1,2\}, \quad i_1 = 2, \quad i_n \neq 1.$$
Set $\mathcal{B} := \{d \text{ in adm. block form}\}$

- Ex.: $1010001 \notin \mathcal{B}$ whereas $101001 \in \mathcal{B}$

- Defn.: Let $\phi: \mathcal{B} \to \{0,1\}^*$, where $\phi(10) := \overline{01}$, $\phi(1001) := \overline{0010}$, and
$$\phi(1 w_{i_1} \cdots w_{i_n} (1 - i_n/2)) := \overline{0 w_{2-i_1} \cdots w_{2-i_n} (i_n/2)}.$$
Say $d \in \mathcal{B}$ satisfies Property $\mathcal{M}$ if $\forall j \geq 0$, $\sigma^j(d) \preceq d$ and $\sigma^j(\overline{\phi(d)}) \preceq d$.
Set $\mathcal{M} := \{d \text{ sat. Prop. } \mathcal{M}\}$

· Prop. 2: $\{\text{matching words}\} \subset \mathcal{M}$

- Sketch: Let d be a matching word.

Write $\begin{pmatrix} d \\ e \end{pmatrix} = \begin{pmatrix} d_1 \cdots d_m \\ e_1 \cdots e_m \end{pmatrix}$, $m > 2$. Recall

$\begin{pmatrix} d_1 \\ e_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} d_{m-2}\, d_{m-1}\, d_m \\ e_{m-2}\, e_{m-1}\, e_m \end{pmatrix} \in \left\{ \begin{pmatrix} 010 \\ 001 \end{pmatrix}, \begin{pmatrix} 001 \\ 010 \end{pmatrix} \right\}$

$= \left\{ \begin{pmatrix} w_2 0 \\ w_0 1 \end{pmatrix}, \begin{pmatrix} w_0 1 \\ w_2 0 \end{pmatrix} \right\}.$

All other digits are determined by cycles at α/β:

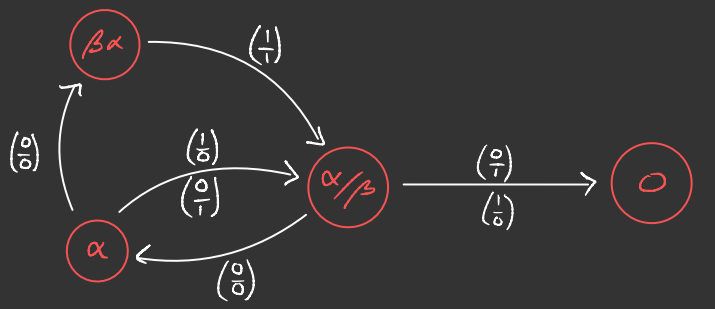
$\begin{pmatrix} d_j\, d_{j+1} \\ e_j\, e_{j+1} \end{pmatrix} \in \left\{ \begin{pmatrix} w_2 \\ w_0 \end{pmatrix}, \begin{pmatrix} w_0 \\ w_2 \end{pmatrix} \right\}$, $\begin{pmatrix} d_j\, d_{j+1}\, d_{j+2} \\ e_j\, e_{j+1}\, e_{j+2} \end{pmatrix} = \begin{pmatrix} w_1 \\ w_1 \end{pmatrix}$.

Thus $d \in \mathcal{B}$ (and $e = \phi(d)$).

Note $x < y$ iff $(S_{\alpha_j}(x)) \prec (S_{\alpha_j}(y))$.

Since $-1 \le S_\alpha^j(1-\alpha)$, $S_\alpha^j(1) \le 1$,

$-d \le \sigma^j(e), \sigma^j(d) \le d \quad \forall j \ge 0.$ □



Vertices: $Y_{j-1} = S_\alpha^{j-1}(1) - S_\alpha^{j-1}(1-\alpha)$

Edges: $\xrightarrow{\binom{d_j}{e_j}}$

Recall: $w_0 = 00$, $w_1 = 001$, $w_2 = 01$, and

$d \in \mathcal{M}$ iff

(i) $d \in \mathcal{B}$, i.e.

$d = 10$, 1001, or

$d = 1\, w_{i_1} \cdots w_{i_n} (1 - i_n/2)$, $i_1 = 2$, $i_n \neq 1$

and

(ii) $\forall j \ge 0$, $\sigma^j(d), \sigma^j(\overline{\phi(d)}) \le d$,

where $\phi(10) = \overline{01}$, $\phi(1001) = \overline{0010}$,

$\phi(1\, w_{i_1} \cdots w_{i_n}(1 - i_n/2)) = \overline{0\, w_{2-i_1} \cdots w_{2-i_n}(i_n/2)}$

- Defn.: Set $I_{10} := (1 + 1/\beta^2, \beta]$, and for $d \in \mathcal{M} \setminus \{10\}$,

$$I_d := \left(\frac{\beta^m + \beta^{d_m}}{\beta^m v(d) + \beta^{d_m}}, \frac{\beta^m - \beta^{1-d_m}}{\beta^m v(d) - \beta^{1-d_m}} \right)$$

- Prop. 3: For each $d \in \mathcal{M}$, $\emptyset \neq I_d \subset [1, \beta]$. Further, for each $\alpha \in I_d$, the S_α-expansions of 1 and $1-\alpha$ begin with d, $\phi(d)$, resp., and $m(\alpha) = \text{len}(d)$.

  - Idea: For $\emptyset \neq I_d \subset [1, \beta]$, compare endpoints explicitly.

  For second statement, again use for $x \in \{1, \alpha - 1\}$, $0 \leq j \leq \ell_\alpha$ that

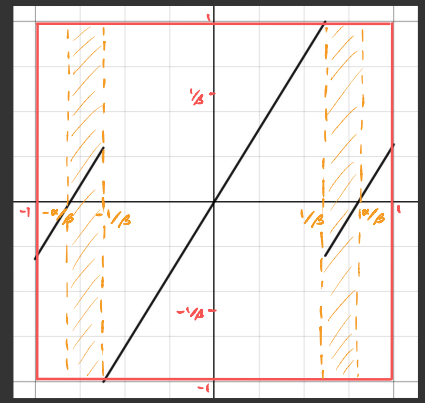
$$S_\alpha^j(x) = \alpha B^j(x/\alpha), \qquad B(x) = \beta x \pmod 1.$$

  Perform careful analysis of B-expansions of $1/\alpha$, $(\alpha-1)/\alpha$.

  That $m(\alpha) = \text{len}(d)$ follows from the final three digits of d, $\phi(d)$ and the orbit-differences graph. □

- Prop's 2, 3 imply $\mathcal{M} = \{\text{matching words}\}$. Further, Prop. 3 implies each $I_d \subset$ 'matching interval corresponding to d'

- Prop. 4: If S_α has matching then $\alpha \in I_d$, where $d \in \mathcal{M}$ is the matching word corresponding to α.



- Idea: Assume $m > 2$. From the orbit-differences graph,

$$\begin{pmatrix} d_{m-2} & d_{m-1} & d_m \\ e_{m-2} & e_{m-1} & e_m \end{pmatrix} \in \left\{ \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \right\}.$$

Suppose $d_m = 0$. Either

$$S_\alpha^{m-1}(1-\alpha) \in (-\alpha/\beta, -1/\beta) \quad \text{or} \quad S_\alpha^{m-2}(1) \in (1/\beta, \alpha/\beta).$$

Use these and $S_\alpha^j(x) = \beta^j(x - \alpha v(s_1 \cdots s_j))$ to obtain bounds on α and show $\alpha \in I_d$. □

- Prop's 2, 3, 4 classify matching words and intervals as $\mathcal{M}$ and $\{I_d\}_{d \in \mathcal{M}}$, respectively

(iii) Invariant measures and frequencies of digits

Invariant measures

- Recall, for $\alpha \in (1, \beta]$ there exists a unique — hence ergodic — absolutely continuous S_α-invariant probability measure μ_α with density

$$f_\alpha(x) = \frac{1}{C} \sum_{j \geq 0}^{m-1} \sum_{a \in \{\pm 1/\beta\}} \frac{1}{\beta^{j+1}} \left(\mathbb{1}_{[-1, S_\alpha^j(\lim_{x \to a^-} S_\alpha(x))]}(x) - \mathbb{1}_{[-1, S_\alpha^j(\lim_{x \to a^+} S_\alpha(x))]}(x) \right)$$

$j \geq 0$ → $j = 0$ when matching occurs

- **Thm. 1:** For each $\alpha \in (1, \beta]$, $\exists!$ abs. cont. S_α-invariant measure μ_α. In fact, $\mu_\alpha \sim \lambda$, and $\exists$ a countable collection $\{I_d\}_{d \in \mathcal{M}}$ of disjoint, open subintervals of $[1, \beta]$ with full λ-measure s.t. for fixed $d \in \mathcal{M}$, the density of each μ_α, $\alpha \in I_d$, is a step function with (at most) the same number of jumps

Frequencies of digits

- By Birkhoff's Ergodic Theorem, for $\alpha \in (1, \beta]$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{1}_{J_i}(S_\alpha^j(x)) = \int_{J_i} d\mu_\alpha = \mu_\alpha(J_i)$$

  for ~~μ_α~~ λ - a.e. $x \in [-1, 1]$

- $1 = \sum_{i=-1}^{1} \mu_\alpha(J_i)$ and $\mu_\alpha(J_{-1}) = \mu_\alpha(J_1)$

- Let $\text{freq}_s : (1, \beta] \to [0, 1]$
  $\alpha \mapsto \mu_\alpha(J_0)$

- Arguments analogous to OK 20 imply continuity

- Since matching α are dense in $(1, \beta]$, consider $\text{freq}_s|_{I_d}$

- For $\alpha \in I_d$, $d \in \mathcal{U}$,

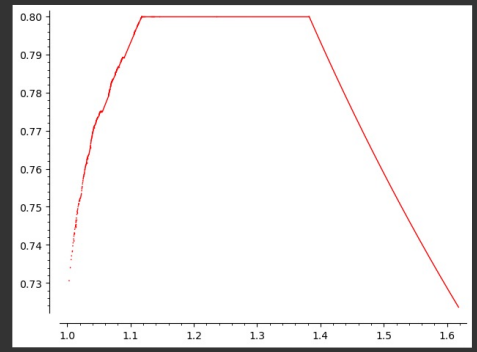
$$freq_s(\alpha) = 1 - \frac{1}{\beta v(\Xi(d))}\left(\frac{n(d)-1}{\alpha} - K(d)\right)$$

where $\Xi(\cdot)$, $n(\cdot)$ and $K(\cdot)$ depend only on d

- $n(d) := \sum_{j=1}^{m} d_j + \sum_{j=1}^{m} e_j$

- Note $freq_s|_{I_d}$ is

$$\begin{cases} \text{strictly increasing}, & n(d) > 1 \\ \text{constant}, & n(d) = 1 \\ \text{strictly decreasing}, & n(d) < 1 \end{cases}$$



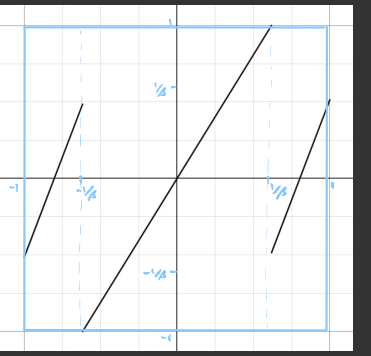
$freq_s(\alpha)$ for $\alpha \in I_d$, $len(d) \leq 20$

- **Thm. 2:** The asymptotic relative frequency of 0 in λ-a.e. S_α-expansion depends continuously on α and attains its max. 4/5 on the maximal interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$

(iv) Jump transformations

- For $\alpha \in [1, \beta]$, define $T_\alpha : [-1, 1] \to [-1, 1]$ by

$$T_\alpha(x) := \beta^{1+|s(x)|} - s(x)\beta x = \begin{cases} \beta^2 x + \beta\alpha, & x \in J_{-1} \\ \beta x, & x \in J_0 \\ \beta^2 x - \beta\alpha, & x \in J_1 \end{cases}$$



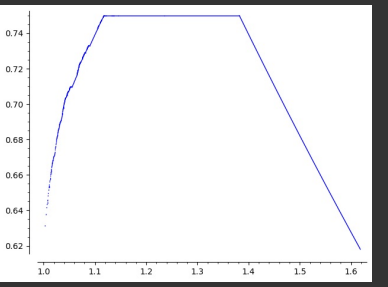
$T_{1.3}$

- T_α is the jump transformation of S_α w/r/t J_0

- Prop.: Each T_α, $\alpha \in (1, \beta]$ has a unique — hence ergodic — T_α-invariant measure ν_α given by

$$\nu_\alpha(A) = \frac{\mu_\alpha(A/\beta)}{\text{freq}_S(\alpha)}.$$

Moreover, $\text{freq}_T(\alpha) := \nu_\alpha(J_0) = 2 - 1/\text{freq}_S(\alpha)$



Analogous results

- Thm. 1: For each $\alpha \in (1, \beta]$, $\exists!$ abs. cont. ~~S_α~~ T_α-invariant measure ~~μ_α~~ ν_α. In fact, ~~μ_α~~ $\nu_\alpha \sim \lambda$, and $\exists$ a countable collection $\{I_d\}_{d \in \mathcal{U}}$ of disjoint, open subintervals of $[1, \beta]$ with full λ-measure s.t. for fixed $d \in \mathcal{U}$, the density of each ~~μ_α~~ ν_α, $\alpha \in I_d$, is a step function with (at most) the same number of jumps

- Thm. 2: The asymptotic relative frequency of 0 in λ-a.e. T_α ~~S_α~~-expansion depends continuously on α and attains its max. ~~4/5~~ 3/4 on the maximal interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$

Open questions

- Is there an explicit (useful) bijection

$$\left\{\begin{matrix}\text{matching words for}\\ \text{symmetric golden maps}\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}\text{matching words for}\\ \text{symmetric binary maps}\end{matrix}\right\} \quad ?$$

- Can we obtain analogous results for symmetric maps of different slopes $m \in (1,2)$,

$$S_{m,\alpha}(x) := \begin{cases} mx+\alpha & , \ x \in [-1, -1/m) \\ mx & , \ x \in [-1/m, 1/m] \\ mx-\alpha & , \ x \in (1/m, 1] \end{cases} \quad , \quad \alpha \in [1, m] \quad ?$$

Thank you!

References

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